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Using Multimodal Large Language Models for Photo Analysis

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Abstract

This thesis investigates the potential of multimodal large language models to be used as a tool for the official automated passport photo classification, an area where high accuracy is demanded. The publicly available MMLLM (Multi-Modal Large Language Model) LLaVa v1.6 Vicuna 13B model was used and adapted using different tactics. The primary objective was to create a model that could classify passport-style photographs according to criteria defined by an international standard. A manually curated dataset was created to serve both as a proof-of-concept and as training material for fine-tuning. In the process of fine-tuning, the learning rate and the number of epochs were systematically varied to achieve the best possible result. The fine-tuned model achieved both F1 and AUC of 0.77. These results demonstrate measurable improvements over the base model. Further improvements are suggested through enhanced dataset diversity and more granular supervision.

Keywords

Multimodal models, LLaVA, photo classification, fine-tuning, QLoRA, document photos.

Abstrakt

Tato bakalářská práce zkoumá potenciál multimodálních velkých jazykových modelů jako nástrojů pro oficiální automatizovanou klasifikaci pasových fotografií, kde se vyžaduje vysoká přesnost. Byl použit veřejně dostupný multimodální model LLaVa v1.6 Vicuna 13B a byl upraven pomocí různých přístupů. Hlavním cílem bylo vytvořit model, který dokáže klasifikovat pasové fotografie podle kritérií definovaných mezinárodní normou. Byla ručně sestavena datová sada na trénování a testování modelu. Během procesu fine-tuningu byly systematicky manipulované hodnoty rychlosti učení a počtu epoch pro dosažení co nejlepšího výsledku. Finálně vyladěný model dosáhl hodnot F1 a AUC rovných 0,77. Tyto výsledky ukazují patrné zlepšení oproti výchozímu modelu, další vylepšení lze dosáhnout zvýšením rozmanitosti datové sady a granularity dohledu nad učním.

Klíčová slova

Multimodální modely, LLaVA, klasifikace fotografií, fine-tuning, QLoRA, fotografie dokumentů.

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Introduction

Recent advances in multimodal large language models (MMLLMs), which seamlessly integrate visual and linguistic capabilities, have opened new pathways for automated image analysis (Carolan et al., 2024). Their wide adoption has made their use for a wide range of tasks easier and cheaper. Simultaneously, there has been a growing demand from government institutions and private enterprises for automated methods to validate photographs for official documents against established standards such as ICAO 9303. Manual review is costly, subjective, and impractical for high-throughput digital services. Thus, a reliable, automated tool for photo compliance checking is highly desirable.

This thesis focuses on evaluating whether it is possible to develop a tool that can effectively classify photos according to some arbitrary list of criteria by adapting open source MMLLM (Multi-Modal Large Language Model), LLaVa v1.6, as well as other publicly available resources (datasets, etc).

First, the thesis investigates the theory behind the multimodal LLMs, explaining how they are different from classical LLMs, how they work by combining inputs from different modes of information, as well as their limitations.

Then, in the practical part, this thesis delves deeper into the specifics behind prompt engineering, model evaluation, dataset creation, and model fine-tuning.

Chapter 2, Prompt engineering, lists different approaches to referring the context of the task to the model, as well as the desired output, then it suggests several versions of the prompt, evaluates them and selects the best to move forward with. Chapter 3, Evaluation methodology, investigates different quantitative metrics that can be collected from the output of the model, and how to use them to grade the model performance, as well as integrating an evaluation mechanism with the model on the code level. Chapter 4, Dataset, investigates the problematics of datasets in the context of model training and testing, the information needed to use the dataset productively and involves the creation of a custom dataset for the task at hand. Chapter 5, Model training, reviews the concept of fine-tuning the model as well as presents specific tools used in the training of the model. Chapter 6, Evaluation, presents the results of evaluation runs of different models.

1 Multimodal Large Language Models

1.1 Overview

MMLLMs (Multi-Modal Large Language Models) are the next evolutionary step of LLMs (Large Language Models) that combine multiple data types, such as images, text, audio, and more (Wu et al., 2023), thanks to which they are able to extract data in a revolutionary fashion, and as such, they allow for a greater amount of complicated tasks to be conducted.

1.2 High-Level Architecture

Upon closer inspection, MMLLM architecture is a combination of a visual system with a language system. Image-based MMLLMs (ratio between the true positive *language models*) are models that “generally consist of a vision encoder, a language encoder, and a cross-modal alignment network” (Carolan et al., 2024).

In plainer terms, such a model has one part that sees the image (the vision encoder) and another part that speaks/understands text (the language model or decoder). Between them, there is usually some mechanism to align the visual information with the text information, so that the two parts can work together.

This is not the only approach to enabling multimodality in LLM, but generally, all the most popular MMLLMs use it. According to a study by Carolan et al., (2024), “Some early approaches consisted of converting an image into an intermediate representation before converting it into a caption. This typically involved using two neural networks, a Convolutional Neural Network (CNN) for the encoder, followed by using the last layer as the input, the feature space, to a Recurrent Neural Network acting as the decoder. But these types of approaches tended to be computationally expensive and produced vague results.” (p. 15)

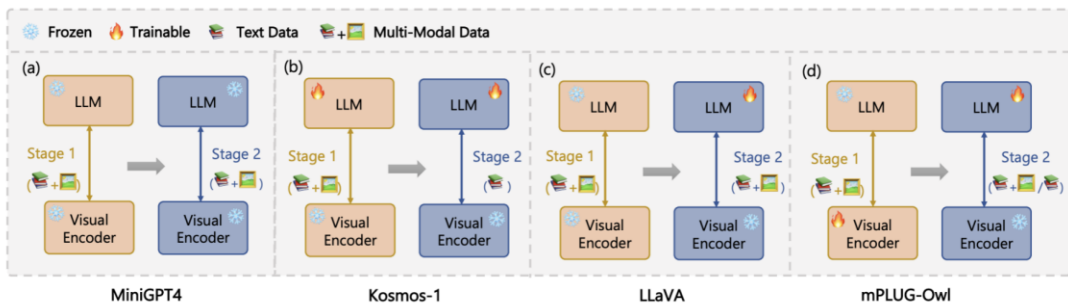


Figure 1: Taxonomy of multimodal LLM training pipelines. Source: Ye et al., 2024

1.3 LLaVA

The focus of this thesis is, specifically, the Large Language and Vision Assistant (LLaVA), which is an MMLLM that introduced a method for connecting a visual encoder and an LLM to produce a model for general-purpose vision and language understanding. It uses CLIP as a visual encoder and Vicuna as the language decoder (model) (Carolan et al., 2024).

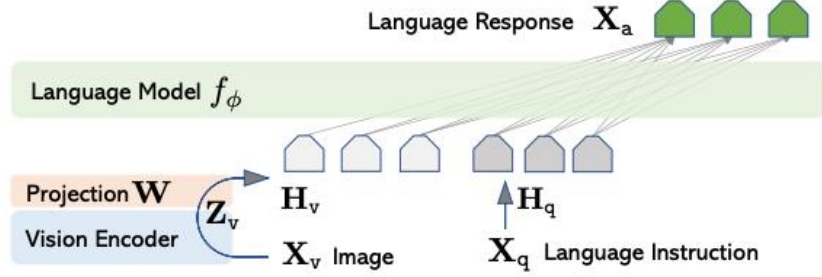


Figure 2: LLaVA network architecture Source: *Llava-Hf/Llava-1.5-13b-Hf · Hugging Face*, 2024

1.3.1 Visual Encoder

CLIP, a visual encoder that is used by LLaVA, is a popular visual encoder used by many multimodal models. It works on the principle of mapping both text and visual information to the same embedding space, which is achieved by using a projection layer that translates the visual information into the lexical space of a language model (*GeeksforGeeks*, n.d.). This allows the language model that works with these embeddings to understand and work with images as well as with text. For example, an image of a blue sky and the sentence “an image of a blue sky” would have very similar embeddings and will be close to each other in a vector space (Palucha, 2024).

1.3.2 Language Model

As mentioned above, LLaVa uses the Vicuna language model, which is a special implementation of LLaMA that was further trained by fine-tuning it on the user conversations from ShareGPT. In performance testing, it showed the results that are close to those of closed-source models like ChatGPT and Google Bard. The whole model (training code, weights and an online demo) is publicly available for non-commercial use (*Vicuna*, n.d.).

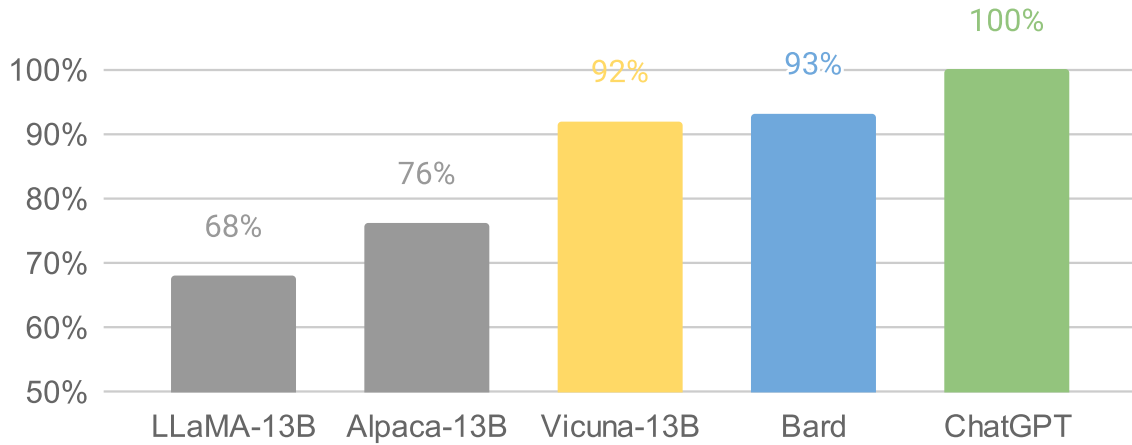


Figure 3: Response quality relative to GPT-4. Source: *Vicuna*, n.d.

1.3.3 Example of the generation workflow

An input image is decoded, centre-cropped to 224×224 px, converted into 14×14 patches and transformed by CLIP into 197 patch embeddings. Those embeddings are then mapped onto a 4096-d “visual token” by the MLP projection layer. That visual token exists in the same embedding space as Vicuna’s lexical tokens. So that, unlike the popular belief, the model doesn’t translate the image to text, it is directly mapped into an intermediate language. In parallel, the user’s text prompt is tokenised by the Vicuna tokenizer. (Liu et al., 2024)

In the process of the answer generation, the Vicuna language model, when predicting the next word attends not only to the previous words but to the “visual token” of the image, allowing visual cues encoded by CLIP, such as color, posture, layout, etc. to influence the next token probability distribution.

1.4 Specific LLaVA version and implementation

As this thesis does not require a specific implementation of LLaVA, the choice of a concrete publicly accessible LLaVA model (of which there are many) was mostly based on the convenience of usage (and accuracy of the output down the line of work).

The model used is the Hugging Face version of *llava-v1.6-vicuna-13b* as accessible on <https://huggingface.co/llava-hf/llava-v1.6-vicuna-13b-hf>. The reason why the Hugging Face version was specifically used is that it is compatible with most of the libraries used throughout the experiment (specifically, the *Transformers* library) (*Llava-Hf/Llava-v1.6-Vicuna-13b-Hf* · Hugging Face, 2024).

Also, an important consideration about this model is that it can only handle one image at a time (Liu et al., 2024).

2 Prompt engineering

One of the tools that will be used to create a reliable model for evaluating photos is **prompt engineering**.

Prompt engineering is the craft of writing the model's input prompt in a way that best elicits the desired output. Since LLaVA is a chatbot that can see images, its prompt can contain several parts, such as user messages and image inputs. The number and style of input parts can be categorised into four different prompting strategies:

- **Zero-shot prompting:** where the model only sees the instruction and the image, without any example of questions and answers (Q&A). This tests the model's ability to follow the instructions purely based on its ability to infer the context from them. It's called zero-shot because the model isn't given any demonstrated example in the prompt about how to do it.
- **One-shot prompting:** where the model sees one example of an image (from a prepared dataset) with its correct label in the prompt. The idea is to give the model a template of response to mimic.
- **Few-shot prompting:** similarly to one-shot prompting, it includes a few different examples (usually 2 or 3) covering different outcomes. The benefit of this strategy is that it shows the model multiple instances of the task, and if the prompt is constructed well enough, the model should be able to generalise the pattern.
- **Chain-of-thought prompting:** This strategy combines the approach of x-shot prompting with explaining the reasoning behind the answer. The example answer in the prompt now contains not only the result but also the reasoning behind the result.

Out of these 4 possibilities, only the **Zero-shot prompting strategy** will be proceeded with, mainly because of LLaVA not supporting more than one image per prompt and because of the difficulties associated with generating the example answer. It is entirely possible to use other strategies through the chat template, but it was decided to manipulate the model further through fine-tuning instead.

Interestingly, large language models can do *in-context learning*, meaning they can pick up the pattern from examples in the prompt without weight updates (Krishnamurthy et al., 2024). That once again suggests the potential effectiveness of more-shot prompt engineering strategies in this task. This strategy should, however, be explored further with a different number of criteria. The reason is that illustrating the nine criteria in the initial examples is quite hard: moreover, if we consider the limitations like prompt length, the decreasing speed of a model, and the 1-image per prompt rule.

Throughout the prompt engineering, the tone of the instructions was kept formal and explicit (because the task is a form of policy compliance checking, where a high precision and strict output format is desired). The general approach was to lay out all the rules with as much clarity and precision in the allowed token length.

2.1 Photo Criteria

The criteria a valid photo must comply with were extracted from the Czech Ministry of Foreign Affairs guidelines for passport photos.

Format	Format (see Photo Template for the size) – proper size is 5 x 5 cm. It also includes image size, head size, head position, and eye position.
Photo Quality	In sharp focus and clear. Neutral colour, natural skin tones, no red eyes. Printed on high-quality paper and at high resolution (min 250 DPI).
Lighting	Appropriate brightness and contrast. Balanced lighting, no shadows or flash reflections on the face.
Eyes	Directly looking at the camera. Eyes open and clearly visible, no hair across the eyes.
Pose	Face must be in the centre. Portrait style and tilted positions are not acceptable. The photograph must show both sides of the face evenly. Please use the template for further checking.
Background	Plain light-coloured (single-coloured) background. The photographed person must be shown alone with a clear background (other persons or items must not be in the picture).
Glasses	Eyes must be shown clearly with no flash reflections on glasses. No tinted glasses. Frames must not cover any part of the eyes. / Avoid heavy frames.
Head Coverings	Head coverings are not permitted except for religious reasons. Facial features from the bottom of the chin to the top of the forehead and both sides of the face must be clearly shown.
Facial Expression	Facial expression must be neutral. The mouth must be closed.

Table 1: Nine official passport photo criteria used for evaluation. Source: Ministerstvo zahraničních věcí České republiky, n.d.

1. Formát fotografie (viz. šablona pro kontrolu velikosti pasové fotografie) – ideální velikost předlohy je 5 x 5 cm. Format (see Photo Template for the size) – proper size is 5 x 5 cm. > Velikost fotografie/ Image size > Velikost hlavy/ Head size > Pozice hlavy/ Head position > Pozice očí/ Eyes position	 ✖ Příliš blízko Too close	 ✖ Příliš vzadu Too far away	 ✖ Příliš malá Too small	 ✔
---	--	---	--	--

Figure 4: Example of valid photo criteria from the document. Source: Ministerstvo zahraničních věcí České republiky, n.d.

The document also includes a more extensive explanation and photo examples, which are also used in crafting the prompt for the former, and in the training and testing datasets for the latter.

2.2 Prompt variants

The base prompt used looked the following way:

```
You are a system that checks if a photo meets the passport photo requirements. Respond in exactly one of two ways:
```

```
'Acceptable'
```

```
'Unacceptable: <One or more criteria>'
```

```
Criteria to consider: Format (head size and position), Photo quality (focus, color, no red-eye), Lighting (brightness, contrast, no shadows), Eyes (open, looking at camera, not occluded), Pose (centered, not tilted, face fully visible), Background (plain light-colored, no other objects), Glasses (allowed only if eyes fully visible, no glare, no tints), Head coverings (none except religious, face fully visible), Facial Expression (neutral, mouth closed).
```

Then the image is provided to the model as an attachment. Notice that the output is constrained heavily, down to the exact words to use. This was intentional to make evaluation easier - it's simpler to check if it said "Acceptable" or not, and to see which criteria were named if unacceptable.

Below are the three prompt variants engineered and sent to the model during the experiments. The wording is kept exactly as the model will see it.

2.2.1 Short

```
You are a system that checks passport photo requirements.
```

```
Analyze the image and respond only with word 'Acceptable' or 'Unacceptable'
```

Token (word) count: 20. The shortest possible instruction. No extra rules, no examples, extreme output cropping. It can show the model's implicit bias as to which photos are acceptable, and which are not.

2.2.2 Medium

```
You are a system that checks if a photo meets passport photo requirements.
```

```
Respond in exactly one of two ways:
```

```
- "Acceptable"
```

- "Unacceptable: <One or more criteria>"

Criteria to consider:

Format: Head size must be just right and be in the centre of the frame

Photo quality: In sharp focus and clear. Neutral colour, natural skin tones, no red eyes.

Lighting: Appropriate brightness and contrast. Balanced lighting, no shadows or flash reflections on face.

Eyes: Directly looking at the camera. Eyes open and clearly visible, no hair across the eyes.

Pose: Face must be in the centre. Portrait style and tilted positions are not acceptable. The photograph must show both sides of the face evenly

Background: Plain light-coloured (single-coloured) background. The photographed person must be shown alone with clear background

Glasses: Eyes must be showed clearly with no flash reflections on glasses. No tinted glasses. Frames must not cover any part of the eyes.

Head coverings: Head coverings are not permitted except for religious reasons. Facial features from bottom of the chin to top of forehead and both sides of the face must be clearly shown.

Facial Expression: Facial expression must be neutral. Mouth must be closed.

Token (word) count: 196. This prompt gives the model the full rule list of photo evaluation criteria, exactly as they were described in the referenced document (see previous chapter). It still, however, demands a very compact answer, only requiring the model to provide a short reason in the case of negative evaluation.

2.2.3 Long

You are a system that checks passport photo requirements.

Evaluate whether the provided photo meets the following passport photo standards:

1. ****Dimensions and Format****:

- Photo dimensions must be 5 cm wide and 5cm high.
- The face must occupy approximately 70-80% of the photo height (31-36mm from chin to top of the head).

2. ****Face Position and Expression****:

- The head must be straight, centered, and facing directly towards the camera.
- Facial expression must be neutral with closed mouth, eyes open and clearly visible.

3. ****Photo Quality****:

- The photo must be sharp, clear, and well-focused.
- The background should be uniform, light-colored (ideally white or light grey), with no patterns or shadows.
- Lighting should be evenly distributed without reflections, shadows, or red-eye.

4. ****Eyes and Glasses****:
 - Eyes must be clearly visible and not obscured by hair or glasses.
 - Glasses are permitted only if lenses are clear and free of glare, reflections, or shadows.
 - Sunglasses or tinted lenses are prohibited.
5. ****Head Coverings and Accessories****:
 - Head coverings are only permitted for religious or medical reasons and must not obscure facial features.
 - No accessories or hair coverings should obscure the outline of the face.
6. ****Clothing and Appearance****:
 - Clothing should contrast clearly with the background.
 - Military, police, or uniformed attire should not be visible.
7. ****Other Considerations****:
 - The photo must represent current appearance, ideally taken within the past 6 months.
 - There must be no visible edits, filters, or digital alterations.

Provide a structured assessment clearly indicating compliance or non-compliance with each guideline.

Here is a template of your response:

This photo is (Acceptable|Unacceptable).

Here is an evaluation of the 7 criteria:

<beforementioned criteria>

Token (word) count: 267. This prompt is attempting to achieve something similar to the chain of thought prompting, forcing the model to explain its reasoning, as well as clearly listing all the criteria with even more precise descriptions. The number of criteria decreased from 9 to 7 because some of them were logically too similar to explain in-depth. It requires the model to produce a complete evaluation of the image against the listed criteria, both in the case of positive and negative evaluations.

3 Evaluation methodology

This chapter contains all the information regarding the criteria of the photo “fitness”, metrics used to evaluate the model's performance in a particular test run, and the evaluation workflow through which each photo is evaluated in each test run.

3.1 Confusion matrix

A confusion matrix is a table used to describe a classification system's performance (in our case, the final decision of the tested model). It summarises and visualises the performance of the model. (*ScienceDirect Topics*, n.d.) The table below shows a confusion matrix of a photo-evaluation algorithm.

Actual	Predicted	
	Acceptable	Unacceptable
Acceptable	TP	FN
Unacceptable	FP	TN

Table 2: Confusion matrix. Source: *ScienceDirect Topics*, n.d.

This matrix consists of four parameters that define the metrics of the classifier. These are:

1. **TP (True Positive):** Represents the number of **acceptable** photos that were evaluated to be **acceptable**.
2. **TN (True Negative):** Represents the number of **unacceptable** photos that were evaluated to be **unacceptable**.
3. **FN (False Negative):** Represents the number of **acceptable** photos that were considered **unacceptable**.
4. **FP (False Positive):** Represents the number of **unacceptable** photos considered **acceptable**.

3.2 Performance Metrics

We will be compiling four basic statistical metrics from the confusion matrix for grading a test run. They will be used as the main attributes for ranking model effectiveness. Each of those metrics shows a different point of view on the effectiveness of a model.

3.2.1 Runtime

The simplest metric showing the overall time the test run took in minutes and seconds.

3.2.2 Accuracy

Accuracy is a metric that evaluates the number of “correct”, meaning true positive or true negative outcomes, against the total number of data points.

$$Accuracy = \frac{TP + TN}{n}$$

Equation 1: The accuracy equation

Accuracy is measured on a scale of 0 to 1. The higher the accuracy, the better. This metric helps to get a quick picture of an overall model performance, especially when the measured classes are balanced (Ohiri, 2024).

3.2.3 Precision

Precision, also known as the positive predictive value (PPV), is a metric that focuses on the accuracy of positive predictions. It measures the proportion of instances the model predicted as positive that were true positives. Mathematically it is defined as a ratio between the true positive outcomes and the true and false positive outcomes (Ohiri, 2024).

$$Precision = \frac{TP}{TP + FP}$$

Equation 2: The precision equation

3.2.4 Recall

“Recall is a classification metric that measures the ability of a model to identify all relevant instances of a particular class. It is defined as the proportion of actual positive instances correctly predicted by the model.” (Ohiri, 2024).

$$Recall = \frac{TP}{TP + FN}$$

Equation 3: The recall equation

It also ranges from 0 to 1, with higher values representing better results. It signifies the ability of a model to correctly identify all the photos that are acceptable.

3.2.5 F1 score

F1 score is also known as the F-Measure. It states the equilibrium between the precision and the recall (*Confusion Matrix - an Overview* | *ScienceDirect Topics*, n.d.).

$$F1\ Score = \frac{2 * precision * recall}{precision + recall}$$

Equation 4: The F1 Score equation

It is a harmonic mean of precision and recall and is a balanced measure of performance.

These metrics were computed using Python's scikit-learn library, as demonstrated below in the excerpt from `utils.py` (Appendix D):

```
# Excerpt from utils.py - metrics calculation
Accuracy = metrics.accuracy_score(actual, predicted)
Precision = metrics.precision_score(actual, predicted,
pos_label=Photo.Class.Acceptable.value)
Recall = metrics.recall_score(actual, predicted, pos_label=Photo.Class.Acceptable.value)
F1_score = metrics.f1_score(actual, predicted, pos_label=Photo.Class.Acceptable.value)
print({"Accuracy":Accuracy, "Precision":Precision, "Recall":Recall, "F1_score":F1_score})
```

3.3 Receiver Operating Characteristic (ROC) Curve

The ROC curve graph represents the relationship between the True Positive Rate (Recall) and False Positive Rate at various threshold settings. It allows for a visual assessment of the model's performance (Fawcett, 2006). The curve itself is complemented by the Area Under the Curve (AUC) metric, where values closer to 1 correspond with better performance.

The ROC curve graph was plotted using Python's matplotlib (Matplotlib — Visualization with Python, n.d.) library, as demonstrated below in the excerpt from `utils.py` (Appendix D):

```
fpr, tpr, _ = metrics.roc_curve(actual_binary, predicted_binary)
roc_auc = metrics.auc(fpr, tpr)

plt.figure()
plt.plot(fpr, tpr, color='blue', lw=2, label='ROC curve (area = %0.2f)' % roc_auc)
plt.plot([0, 1], [0, 1], color='gray', lw=2, linestyle='--')
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('Receiver Operating Characteristic')
plt.legend(loc="lower right")
plt.grid()
plt.show()
```

3.4 Graphical performance representation (GPR)

For an enhanced overview of the test runs, an additional graphical performance representation method was introduced. It is a visualisation involving a grid of colored squares representing each individual photo. The colours correspond to the confusion matrix categories (TP, FP, TN, FN):

- Green: True Positive (TP)
- Red: False Positive (FP)
- Blue: True Negative (TN)
- Orange: False Negative (FN)

This visualisation is particularly useful for rapid debugging and identification of specific problematic cases in model predictions. It provides more data on how each photo performed.

4 Dataset

Arguably, one of the most essential things for making a reliable model is having a diverse dataset for training. The task at hand demands a dataset of photos differentiated by the **nine evaluation criteria** (Table 1). Which means that the dataset should contain enough photos so that:

- All the criteria are represented.
- Considering the cases where multiple criteria signal the photo's unfitness, a sufficient number of permutations of these criteria should be present within the unacceptable photos.
- Edge cases should be present. So, when the photo is comparatively slightly unacceptable, the model should be able to detect it.
- Ideally, similar photos should be present within the acceptable and unacceptable categories.

Another crucial part of the dataset is the text descriptions of the photos, both acceptable and unacceptable. Ideally, each image should have a complete textual description of how it grades on every criterion. This is especially needed in the training stage, where the construction of an example training output message must be facilitated by these textual descriptions. To recapitulate, a dataset must:

1. Have at least a thousand photos with a 1:1 ratio of acceptable vs unacceptable.
2. Have every image labelled as acceptable or unacceptable, along with a further description of the nine criteria in the case that it is not acceptable.
3. Be diverse by subject in terms of identities, ethnicities, gender identities and age brackets.
4. Contain multiple types of photos, such as diverse types of mugshots as well as non-formal photos.
5. Have high-quality data.

After conducting a search of a dataset that complies with at least some of the criteria, no such dataset was found in the public domain. This means that the desired dataset for the training of this model must be created manually.

Because no public collection satisfied the nine evaluation (Table 1), an ad-hoc pilot corpus was assembled as a dataset for the purpose of testing the bare/fine-tuned model, as well as facilitating proof-of-concept training runs.

The target was a small but balanced dataset that:

- Covered every criterion at least once
- Contains qualitatively similar acceptable and unacceptable example photos
- Includes a short description for each negative case

As this is the first run, its goal was mainly to check the suggested workflow, without the ambition of creating a production-ready dataset.

4.1 Source selection and filtering

Four publicly available repositories were considered for datamining:

Dataset name	Source	License information
BioID Face Database	<i>The BioID Face Database BioID, 2022</i>	<i>Free to use for face-detection research</i>
Selfie & Official ID Photo Dataset by Axon labs	<i>Selfie & Official ID Photo Dataset (18k+ Images), n.d.)</i>	Creative Commons CC BY 4.0.
ONOT Mugshot dataset	<i>ONOT, n.d.</i>	Creative Commons CC BY-NC 4.0
Photos extracted from the document listing the photo criteria	Ministerstvo zahraničních věcí České republiky, n.d.	Exempt from copyright protection under the Czech law.

Table 3: Public datasets used. Source: Author

The BioID Face Database was discarded because it only contains monochrome photos; this could have had unexpected results after training the model on a mix of colored and monochrome photos (considering the presumption that monochrome photos are not allowed for usage in official documents). The remaining sources yielded 194 photographs (of which 92 were labelled Acceptable and 102 labelled Unacceptable).

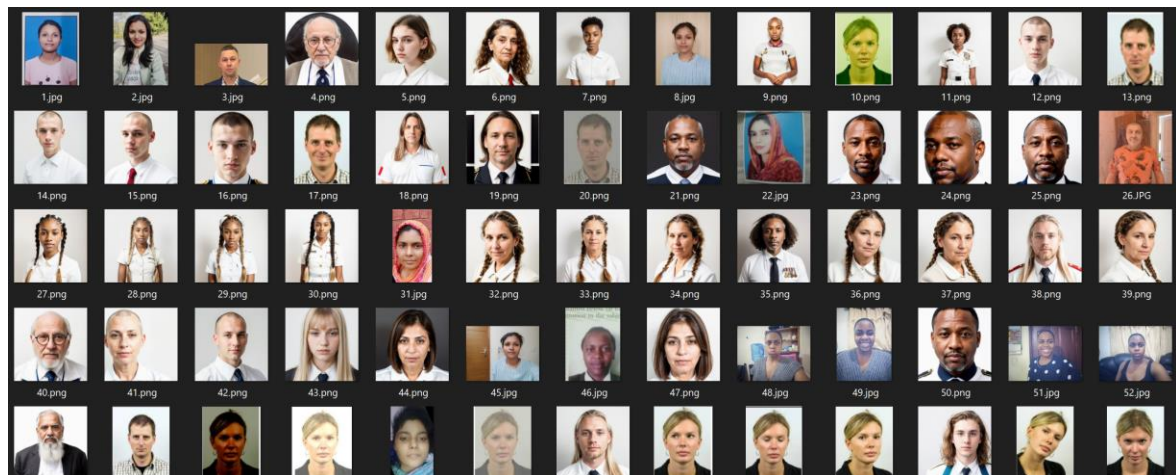


Figure 5: An example of photos from the dataset classified as „Unacceptable“. Source: author

4.2 Captioning the reasons

For each image, labels were assigned through manual inspection. Acceptable images were marked clearly as such, while unacceptable images received a short 1-3 sentence description label identifying the most out-of-order feature in the photo. Captions were kept concise to bring attention to the most prominent feature that a photo demonstrates. For example:



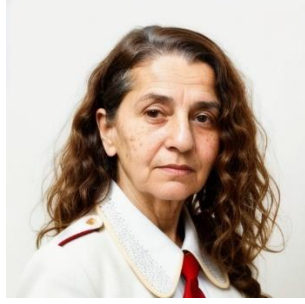
Background: not uniform, the person is not alone in the frame.
Format: head size too big and off-centre.



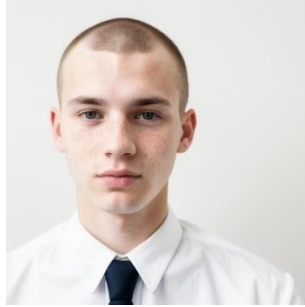
Lighting: shadows on the face.



Photo quality: Unnatural color tint (yellowish/greenish).



Pose: tilted, not directly facing camera. Lighting: shadow on face.



Format: Head is not centered.

Table 4: Several photos classified as unacceptable. Source: Author.

Captions were saved in a .txt file where the line number corresponded to the photo with the same number in the name.

4.3 Annotation protocol and validation

All 194 images in the dataset were labelled exclusively by the author; to mitigate intra-rater bias, the annotation was conducted in two passes separated by a week-long “cool-down” period. During the second pass, any labels that differed from the initial annotation were re-examined against the nine evaluation criteria (Table 1). After the two manual passes, a third one was done with the help of the untrained model. After these passes, the dataset contained 76 acceptable and 118 unacceptable pictures; the imbalance in these classes of photos needed to be addressed to avoid bias towards one of them, so 42 of the unacceptable photos were removed via random selection. The final dataset after the balancing contained only 152 (76 acceptable and 76 unacceptable) photos.

4.4 Transforming

The created data was transformed to a format appropriate to programmable data extraction. The JSON file format was chosen, with a single file that contains metadata of every photo, together with the instructions and desired caption. This transformation is achieved through a simple Python script that takes every reason for unacceptable photos, connects them with actual photos and instructions, and the same goes for the acceptable photos, but in that case, no reason is provided, and the sample output is just the label “Acceptable”. Unacceptable and acceptable photos were sorted into two folders for easy work with the dataset.

```

{
  "image": "/acceptable/1.png",
  "instruction": "\n          You are a system that checks if a photo
meets passport photo requirements.\n          Respond in exactly one of two
ways:\n          - \"Acceptable\"\n          - \"Unacceptable:
<One or more criteria>\"\n          Criteria to consider:\n
Format: Head size must be just right and be in the centre of the frame\n
Photo quality: In sharp focus and clear. Neutral colour, natural skin tones, no red
eyes.\n          Lighting: Appropriate brightness and contrast. Balanced
lighting, no shadows or flash reflections on face.\n          Eyes: Directly
looking at the camera. Eyes open and clearly visible, no hair across the eyes.\n
Pose: Face must be in the centre. Portrait style and tilted positions are not acceptable.
The photograph must show both sides of the face evenly\n
Background: Plain light-coloured (single-coloured) background. The photographed person
must be shown alone with clear background\n          Glasses: Eyes must be
showed clearly with no flash reflections on glasses. No tinted glasses. Frames must not
cover any part of the eyes.\n          Head coverings: Head coverings are
not permitted except for religious reasons. Facial features from bottom of the chin to top
of forehead and both sides of the face must be clearly shown.\n
Facial Expression: Facial expression must be neutral. Mouth must be closed.",
  "output": "Acceptable."
},
{
  "image": "/acceptable/10.png",
  "instruction": "\n          You are a system that checks if a photo
meets passport photo requirements.\n          Respond in exactly one of two
ways:\n          - \"Acceptable\"\n          - \"Unacceptable:
<One or more criteria>\"\n          Criteria to consider:\n
Format: Head size must be just right and be in the centre of the frame\n
Photo quality: In sharp focus and clear. Neutral colour, natural skin tones, no red
eyes.\n          Lighting: Appropriate brightness and contrast. Balanced
lighting, no shadows or flash reflections on face.\n          Eyes: Directly
looking at the camera. Eyes open and clearly visible, no hair across the eyes.\n
Pose: Face must be in the centre. Portrait style and tilted positions are not acceptable.
The photograph must show both sides of the face evenly\n
Background: Plain light-coloured (single-coloured) background. The photographed person
must be shown alone with clear background\n          Glasses: Eyes must be
showed clearly with no flash reflections on glasses. No tinted glasses. Frames must not
cover any part of the eyes.\n          Head coverings: Head coverings are
not permitted except for religious reasons. Facial features from bottom of the chin to top
of forehead and both sides of the face must be clearly shown.\n
Facial Expression: Facial expression must be neutral. Mouth must be closed.",
  "output": "Acceptable."
}

```

```

},
{
  "image": "/unacceptable/30.png",
  "instruction": "\n
You are a system that checks if a photo
meets passport photo requirements.\n
Respond in exactly one of two
ways:\n
- \"Acceptable\"\n
- \"Unacceptable:
<One or more criteria>\"\n\n
Criteria to consider:\n
Format: Head size must be just right and be in the centre of the frame\n
Photo quality: In sharp focus and clear. Neutral colour, natural skin tones, no red
eyes.\n
Lighting: Appropriate brightness and contrast. Balanced
lighting, no shadows or flash reflections on face.\n
Eyes: Directly
looking at the camera. Eyes open and clearly visible, no hair across the eyes.\n
Pose: Face must be in the centre. Portrait style and tilted positions are not acceptable.
The photograph must show both sides of the face evenly\n
Background: Plain light-coloured (single-coloured) background. The photographed person
must be shown alone with clear background\n
Glasses: Eyes must be
showed clearly with no flash reflections on glasses. No tinted glasses. Frames must not
cover any part of the eyes.\n
Head coverings: Head coverings are
not permitted except for religious reasons. Facial features from bottom of the chin to top
of forehead and both sides of the face must be clearly shown.\n
Facial Expression: Facial expression must be neutral. Mouth must be closed.",
  "output": "Unacceptable. Format: head size too big."
}

```

In the final preprocessing step, the dataset.json was converted further, into parquet file format for ease of work in the training process.

The final folder structure of a dataset looks the following way:

- dataset.json
- dataset.parquet
- acceptable
 - 1.jpg
 - 5.jpg
 -
 - 76.jpg
- unacceptable
 - 1.jpg
 - 2.jpg
 - ...
 - 117.jpg

It was ensured that the label wording in the output matches the desired model output. For instance, if an image showed a person with closed eyes, the output label was always phrased as "Eyes: eyes closed.", or similar text with minimal differences, to keep a uniform style.

This consistency helps the model learn the pattern more easily than if one label said, “eyes closed,” and another said, “the eyes are not open”, etc.

This dataset then has been split into two parts: the train and test datasets, with the test part containing 30 images (15 acceptable and 15 unacceptable) and the train part containing 122 (61 acceptable and 61 unacceptable) images.

4.5 Limitations

While the pilot corpus proved useful for a proof-of-concept run, several weaknesses beyond its small size constrain its evidential value:

- **Sampling bias.** Images were drawn from 3 datasets that were not designed for this application. This leads to an under-representation of everyday lighting conditions, modern smartphones, and certain demographics (children, older adults and diverse skin-tones), potentially limiting generalisability.
- **Uneven criterion coverage.** Although every ICAO-derived rule is present at least once, some of them occur only in combination with others (e.g. head coverings almost always co-occur with eyes partly obscured). Single-criterion violations, such as tinted glasses or extreme background glare, are not present in the samples.
- **Annotation subjectivity.** Labels were produced by a single annotator; subtle cases, such as slightly tilted heads or marginal crops, may therefore reflect the annotator’s personal threshold rather than an objective rule.
- **Limited permutation space.** With 9 criteria, there are $2^9 - 1$ possible unacceptable combinations; the corpus contains only a tiny fraction of these. This limits the model’s ability to learn more complex multi-criterion failures.

5 Model training

While prompt engineering can be an effective strategy in wielding MMLLMs, it cannot sufficiently address the fundamental limitations in the model’s capabilities. To mitigate these limitations, supervised fine-tuning was performed. Fine-tuning is a training method in machine learning, where the model parameters are iteratively updated on a prepared training dataset. With this method, the model can learn patterns that are directly associated with the task at hand, improving its predictive accuracy and reliability (Howard & Ruder, 2018).

5.1 QLoRA Overview

Considering the substantial computational requirements associated with training a large-scale multimodal model, such as one used throughout this thesis (LLaVA v1.6 with 13b parameters), it was necessary to employ a more resource-efficient method. The Low-Rank Adaptation (LoRA) method was chosen to be used in this thesis. It works by introducing a small additional parameter layer, which will be updated instead of retraining all the model weights (Hu et al., 2021).

This approach reduces both computational load and memory usage significantly, making the fine-tuning of large models possible even on a single high-end graphics card. In spite of the fact that only a small amount of weights is being updated, the LoRA fine-tuning method demonstrated a performance close to that of traditional full fine-tuning, while requiring fewer computational resources (Hu et al., 2021).

Even though the described method promises superior performance and availability, it would not be possible to use Low-Rank Adaptation as is. The reason being that the training process of 13b parameter model with 16-bit precision would require around 52 GB of GPU memory, when only a maximum of 46 GB of GPU memory is available in the computer used for the writing of this paper. Hence, an adoption of QLoRA (Quantized Low-Rank Adaptation) was required. QLoRA is an extension of LoRA that combines its approach of detaching trained weights with 4-bit normal float quantization, which is a method designed to optimally quantize the weights of neural networks into a 4-bit format (Amur, 2023).

5.2 Training methodology

The fine-tuning process implemented in this thesis can be viewed as a standard supervised learning approach. Each training example consisted of three elements: an input image, a text-based instruction prompt describing the evaluation criteria (described previously in Chapter 4), and the corresponding labelled output (ground truth) indicating whether the image is "Acceptable" or "Unacceptable," alongside relevant criteria if applicable.

The fine-tuning process was designed to minimise the difference between the model's generated textual output and the ground truth annotations. Throughout multiple epochs of training, the model's parameters were iteratively adjusted using gradient-based optimisation algorithm provided by PyTorch (Paszke et al., 2019).

The optimisation was carried out with the AdamW algorithm. AdamW is an optimisation algorithm that adapts the learning rate for each input parameter during training. It achieves this by combining the Momentum and RMSprop gradient techniques. It is used with its default input parameters $\text{Beta1} = 0.9$ and $\text{Beta2} = 0.999$, that are the decay rates for moving averages of the gradient (Momentum) and squared gradient (RMSprop). The initial learning rate was treated as a hyperparameter was explored in the range $1e-6 - 2e-5$.

For image processing, each input photo was first transformed into embeddings using the CLIP (Contrastive Language-Image Pre-training) visual encoder. These embeddings were subsequently passed, together with the textual prompts, to the Vicuna-based text decoder, responsible for generating textual predictions.

The training process did not employ reinforcement learning or any advanced optimisation methods; instead, a straightforward supervised fine-tuning pipeline was utilised.

5.3 Hyperparameter Tuning Strategy

Fine-tuning performance significantly depends on hyperparameter selection. Hu et al. (2021) indicate that learning rate, number of epochs, and batch size are critical hyperparameters affecting LoRA fine-tuning outcomes. In this thesis, two key hyperparameters were systematically explored:

- **Learning rate (lr):** Adjustments to this parameter directly control the size of model parameter updates during training. An excessively high learning rate may cause unstable learning and overfitting, while an overly low rate can prevent meaningful convergence (Hu et al., 2021)
- **Number of epochs:** This parameter determines how many times the model iterates through the training dataset. Higher numbers of epochs allow deeper learning of training patterns, but also increase the risk of overfitting, especially with smaller datasets (Hu et al., 2021).

Due to computational limitations, **batch size** tuning was not conducted and remained at 1 throughout the whole setup. In their research, Hu et al. (2021) mention that the tweaking of this parameter does not have a significant effect on the result of training.

Hyperparameter tuning was performed iteratively, starting with a default configuration, followed by incremental adjustments based on the model's evaluation outcomes. Every try started with the learning rate of $1e-5$, and then every consecutive try was performed with a different learning rate, different by some factor (usually 2 or 0.5). After assessing the results, other base rates and ranges were explored. The same process was performed later with a different epoch number.

The training script was adopted from a publicly available tutorial (*NielsRogge*, n.d.). And was further edited to better fit the needs of the task at hand. The edited version is accessible in Appendix B.

The prompt used was the “medium length” prompt from the prompt engineering chapter as it showed the best overall results in testing.

6 Evaluation

6.1 Evaluation workflow

The evaluation workflow that gets applied to every photo in the evaluation process consists of the following steps:

1. Applying the chat template to the image and prompt.
2. Processing the resulting chat into the model input.
3. Referring the input to the model to receive its evaluation of a photo.
4. Decoding and extracting the label

It can be observed in this exact order in the excerpt from `LlavaModelNext.py` (Appendix D):

```
messages = [
    {
        "role": "user",
        "content": [
            {"type": "image", "image": image},
            {"type": "text", "text": prompt},
        ],
    }
]

text = self.processor.apply_chat_template(
    messages, tokenize=False, add_generation_prompt=True
)

inputs = self.processor(
    text=text,
    images=image,
    return_tensors="pt",
).to("cuda")

outputs = self.model.generate(**inputs, max_new_tokens=512)
response_full = self.tokenizer.decode(outputs[0], skip_special_tokens=True)

assistant_marker = "ASSISTANT:"
if assistant_marker in response_full:
    response = response_full.split(assistant_marker)[-1].strip()
else:
    response = response_full.strip()
return response
```


When all the photos are put through this workflow, the aggregated results are then evaluated, producing different metrics to be used as performance indicators.

6.2 Prompt engineering

	AUC	Accuracy	Precision	Recall	F1_score	Total run time
Bare model short prompt	0.63	0.63	0.58	1.00	0.73	0:17
Bare model medium prompt	0.73	0.73	0.77	0.67	0.71	0:46
Bare model long prompt	0.60	0.60	0.80	0.27	0.40	8:18

Table 5: Test results for the prompt engineering evaluation runs. Source: author

The test dataset used for this evaluation contained 30 images (15 acceptable and 15 unacceptable).

As is visible, the **short prompt length** results in the model correctly classifying every acceptable photo (perfect recall), but it comes at the cost of many false positives. The strategy of the model here is to mark the photo as acceptable unless it sees a substantial reason for it to be classified as unacceptable. And in this case, the result of the evaluation highly depends on the data the model was originally trained on (be it the visual encoder or the language model).

The **long prompt length** triggers the model to be very cautious, labelling far fewer images as positives (high precision), but at the same time, it misses most true positives. In general, a sharp performance drop is observed. It can be speculated that, when the model is presented with such a high number of criteria, together with a specific description to each of them, it begins looking for every reason that the image might be classified as unacceptable, inevitably hallucinating some of the evaluation results. This, combined with a high number of criteria, creates multiple points of failure in the evaluation, which may explain the observed performance drop relative to other prompt variants.

The **medium prompt length** serves as an acceptable middle ground between these two extremes. It does not leave the model “on its own” in the reasoning process, simultaneously not polluting it with extra data that may lead to hallucinations. It has the highest overall AUC and the best accuracy, while it has average or “close to the best” performance on all other metrics. All of this can be considered a satisfactory result at this stage of the model

creation process, and for simplicity, only the medium prompt length will be used for the further steps.

6.2.1 Example outputs

Short

Acceptable

Medium

Unacceptable: Eyes are not directly looking at the camera.

Long

This photo is Unacceptable.

1. Dimensions and Format: The photo dimensions are 5cm wide and 5cm high, which is within the passport photo standards. However, the face does not occupy approximately 70-80% of the photo height, as it is slightly smaller than the required range.
2. Face Position and Expression: The head is straight and centered, but it is not facing directly towards the camera. The face is not fully visible due to the angle of the photo.
3. Photo Quality: The photo is not sharp, clear, or well-focused. The background is not uniform, light-colored, or free of shadows.
4. Eyes and Glasses: The eyes are not clearly visible, as they are partially obscured by the hair.
5. Head Coverings and Accessories: There are no head coverings or accessories visible in the photo.
6. Clothing and Appearance: The clothing contrasts clearly with the background, and there are no visible edits, filters, or digital alterations.
7. Other Considerations: The photo appears to be taken within the past 6 months, and there are no visible edits or filters. However, the photo does not represent current appearance, as the person's hair is styled differently from the photo.

Please note that this assessment is based on the information provided and does not take into account any additional context or requirements that may not be visible in the image.

6.2.2 Other metrics

Other metrics, such as confusion matrix, Receiver Operating Characteristic (ROC) and Graphical Performance representation (GPR) for every evaluation run are displayed in the following table.

	Confusion matrix	ROC	GPR
Short prompt			
Medium prompt			
Long prompt			

Table 6: Confusion matrix, Receiver Operating Characteristic, Graphical Performance representation for the prompt engineering evaluation runs. Source: author

6.3 Fine tuning

	AUC	Accuracy	Precision	Recall	F1_score	Total run time
1 epoch, lr 2e-5	0.67	0.67	0.60	1.00	0.75	1:08
1 epoch, lr 9e-6	0.77	0.77	0.79	0.73	0.76	2:06
1 epoch, lr 7e-6	0.73	0.73	0.73	0.73	0.73	1:47

3 epochs, lr 1e-5	0.53	0.56	0.52	1.00	0.68	0:33
3 epochs, lr 5e-6	0.70	0.70	0.64	0.93	0.76	0:46
3 epochs, lr 2e-6	0.77	0.77	0.75	0.80	0.77	1:05
3 epochs, lr 1e-6	0.77	0.77	0.79	0.73	0.76	1:03

Table 7: Tests result for the fine-tuning evaluation runs. Source: author

The results of testing the fine-tuning on the dataset are quite promising, with AUC approaching the 0.8 figure. It should still be taken with a grain of salt, considering the small size and diversity of the dataset. It is interesting that the models “1 epoch, lr 9e-6” and “3 epochs, lr 1e-6” showed the same result, but this can be attributed to the small size of the training dataset.

6.3.1 Example Outputs

The outputs of all the models trained were not as varied as in the evaluation of different prompt lengths and were similar to one of the two following examples:

Acceptable

Unacceptable: Eyes are not directly looking at the camera.

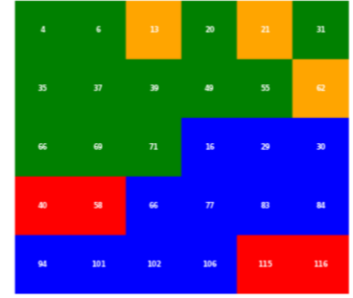
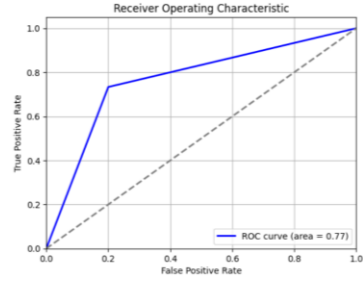
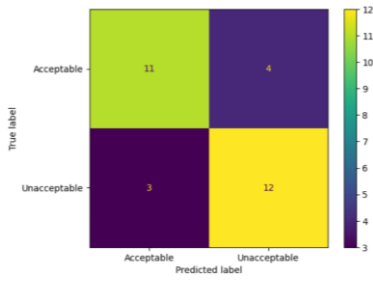
With the latter one mentioning the relevant issue of a specific photo.

6.3.2 Other metrics

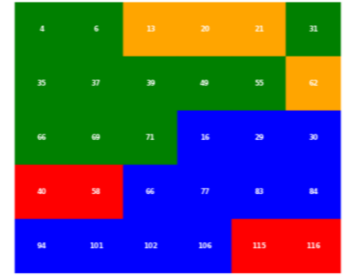
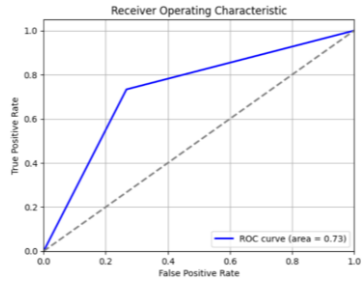
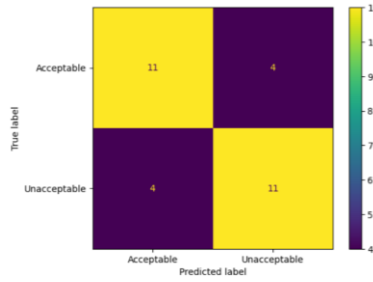
Other metrics, such as confusion matrix, Receiver Operating Characteristic (ROC) and Graphical Performance representation (GPR) for every evaluation run are displayed in the following table.

	Confusion matrix	ROC curve	GPR
1 epoch, lr = 2e-5	Confusion matrix for 1 epoch, lr = 2e-5. The matrix shows 15 True Acceptables, 0 False Acceptables, 10 True Unacceptables, and 5 False Unacceptables.	ROC curve for 1 epoch, lr = 2e-5. The curve shows a True Positive Rate of approximately 0.67 and a False Positive Rate of approximately 0.67.	GPR for 1 epoch, lr = 2e-5. The GPR shows a distribution of scores across the range of 0 to 1.0, with a peak around 0.67.

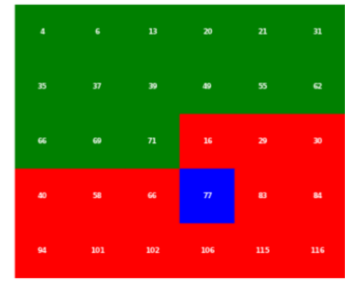
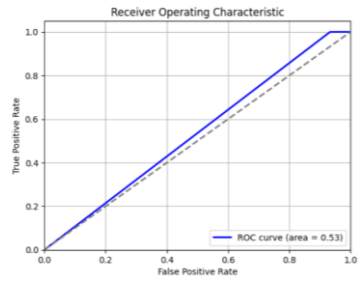
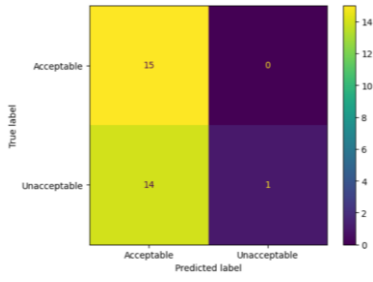
1 epoch, lr = 9e-6



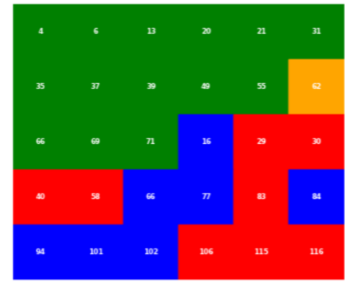
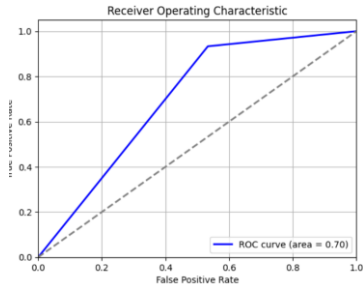
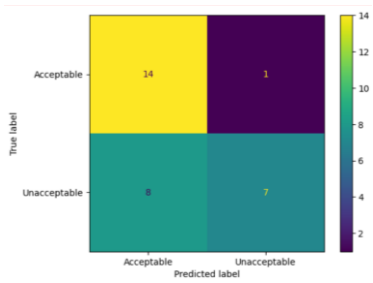
1 epoch, lr = 7e-6



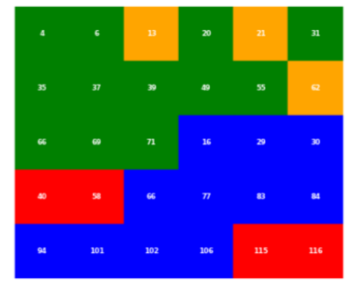
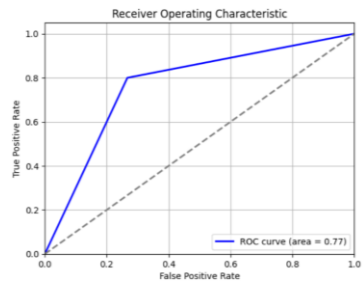
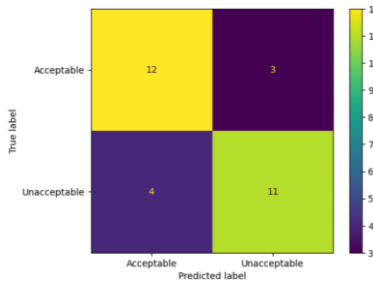
3 epochs, lr = 1e-5



3 epochs, lr = 5e-6



3 epochs, lr = 2e-6



3 epochs, lr = 1e-6

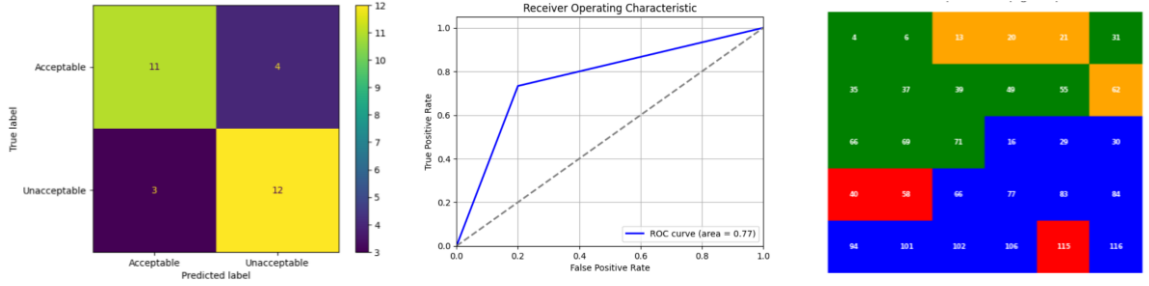


Table 8: Confusion matrix, Receiver Operating Characteristic, Graphical Performance representation for the fine-tuning evaluation runs. Source: author

Result Discussion

The fine-tuning experiments on the 152-image dataset produced an AUC of 0.77 and an accuracy of 0.77 under the best hyperparameter setting (3 epochs, learning rate of $2e-6$). Although the absolute gain over the “untrained” baseline (AUC 0.73; accuracy 0.73) is modest, it is meaningful given the small evaluation set ($n = 30$) and confirms that QLoRA is able to fine-tune the standard model within the use case and with data provided. Learning rate influenced performance more than epoch count. Tripling the number of epochs from 1 to 3 without lowering lr caused the AUC to fall from 0.67 to 0.53, indicating possible over-fitting. When lr was reduced to $2e-6$, the three-epoch run recovered an AUC of 0.77, matching the single-epoch optimum. This pattern is consistent with prior reports that lower learning rates are required when the effective dataset size is small (Hu et al., 2021).

Precision and recall peaked at 0.79 and 0.80 (excluding the overall “bad” result with a recall of 1.00), respectively, with both test cases yielding an F1 of 0.77. False-negative and false-positive errors were largely confined to borderline cases such as slight head-tilt or faint shadows, suggesting that these visual cues remain ambiguous in the current training set.

With fewer than 200 unique identities, the corpus risks model memorisation and demographic bias. Moreover, because every unacceptable image is accompanied by only a short free-text explanation, the model receives little supervised signal about which criterion is violated, or what characteristic is being looked for in the image overall; this may cap the attainable accuracy. Despite these constraints, reaching ~ 0.8 AUC demonstrates that MLLMs can learn the coarse concept from very limited data. However, deployment-grade performance (> 0.9 AUC) will almost certainly require a larger, professionally annotated corpus with much wider coverage of different situations.

Limitations and future work

The model is prompted to provide reasons, but the correctness of those reasons is not formally measured. While the model generates reasons, the study's formal evaluation framework does not score their accuracy.

Another limitation of the evaluation setup is the absence of cross-validation, which is typically advisable when working with small datasets. Due to the limited size of the dataset (152 images), using a single train-test split may not provide a fully reliable estimate of model generalisation. A k -fold cross-validation (5 or 10-fold) would have allowed a more robust assessment of model performance by reducing the variance associated with a single split. This improvement is recommended for future experiments.

Addressing the limitations of the dataset described in Chapter 4, a further priority is the revamping of the dataset. Collecting an additional 2,000–3,000 annotated images with balanced demographic representation across age, gender, and ethnicity would significantly

improve the model's generalisation and fairness. Moreover, transitioning from a binary classification output (“Acceptable” vs “Unacceptable”) to a multi-label system, where each of the defined photo criteria is individually assessed, could enable the model to learn more granular distinctions and enable it to localise specific compliance violations with higher accuracy. With a larger and more diverse dataset, it would also be feasible to incorporate advanced training strategies such as hard-negative mining, which focuses training on challenging misclassified examples, and consistency regularisation, which encourages the model to produce stable outputs for similar inputs.

Conclusion

The main goal of this thesis was to investigate the possibility of using multimodal Large Language Models for photo analysis. Over the course of this work, a few options of adapting the model for this use case were investigated.

The main takeaway is that the creation of a basic model with moderate performance can be achieved casually through the means of prompt engineering. The basic setup with three prompts under the evaluation yielded the prompt that produces an AUC of 0.73 and an accuracy of 0.73 on the test dataset.

Considering the demand for a model that can perform with a high degree of accuracy, the fine-tuning shows promising results, improving AUC and accuracy by 4%, moving the figures closer to the 80-percent mark. This cannot be considered a substantial increase, but it is an admissible and reasonable result that proves the feasibility and impact of fine-tuning in the task of feature extraction.

The principal limitation of the present study is the modest size and diversity of the available dataset. A substantially larger, professionally annotated corpus would almost certainly permit further gains in accuracy beyond the current ~5 percentage-point improvement achieved through fine-tuning. Future work should therefore prioritise expanding the dataset, particularly to cover demographic and illumination conditions that are currently under-represented.

Also, initially it was decided that for the sake of simplicity, the model would be grading photos in a binary fashion, which limits the informational value of this classification. It could be speculated that if the model had more freedom in the output and the resources to base its evaluation on, it could have made a better classification engine.

Regardless of the obstacles encountered, the conducted work resulted in a working basic model performing at a good level, it also laid a clear foundation for further endeavors in this area, so that, under the condition that more resources would be available, a better model could be trained based on the developed code base.

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Appendices

Appendix A: Prompt evaluation

File *prompt-evaluate.ipynb* contains the code and the results of different prompt evaluations. The file is available at <https://github.com/miahandrio/BP-sources>.

Appendix B: Fine tuning

File *fine-tune.ipynb* contains the code for fine-tuning a model using QLoRA. The file is available at <https://github.com/miahandrio/BP-sources>.

Appendix C: Fine-tuned model evaluation

File *fine-tuned-evaluate.ipynb* contains the code and the results of different fine-tuned model evaluations. The file is available at <https://github.com/miahandrio/BP-sources>.

Appendix D: Model and utility files

Files *LlavaNextFineTuned.py*, *LlavaNextModel.py* contain the code for accessing the model. Files *photo.py*, *utils.py* contain utility logic that was used to simplify some code fragments. The files are available at <https://github.com/miahandrio/BP-sources>.

Appendix E: Dataset

The images used for training and testing, together with their annotations and dataset files, are available at <https://github.com/miahandrio/BP-sources>.